Question 3

(a) estimate_kernel_a.py- Instead of exact probabilities, we estimate:

$$\hat{P}((s,a),s') = \frac{\text{Number of times next state is } s'}{K}$$

where: K= number of simulator samples per (s, a)

Working- For each state-action pair (s, a):

1. Run simulator K times.

2. Record resulting next states

2. Count occurrences of each next state

3. Normalize counts to obtain probabilities

(b) run_experiment_q3.py-

Use estimated kernel $\hat{P}$ instead of exact P: $Q_{\text{est}} = VI(\hat{P}, R)$

Comparison

We compare with: $Q_{\text{exact}} = VI(P, R)$

Metric: L1 Difference $L1 = \sum_{s,a} | Q_{\text{exact}}(s,a) - Q_{\text{est}}(s,a) |$

Working- For different values of K:

1. Estimate kernel $\hat{P}$
2. Run Value Iteration
3. Compute L1 difference

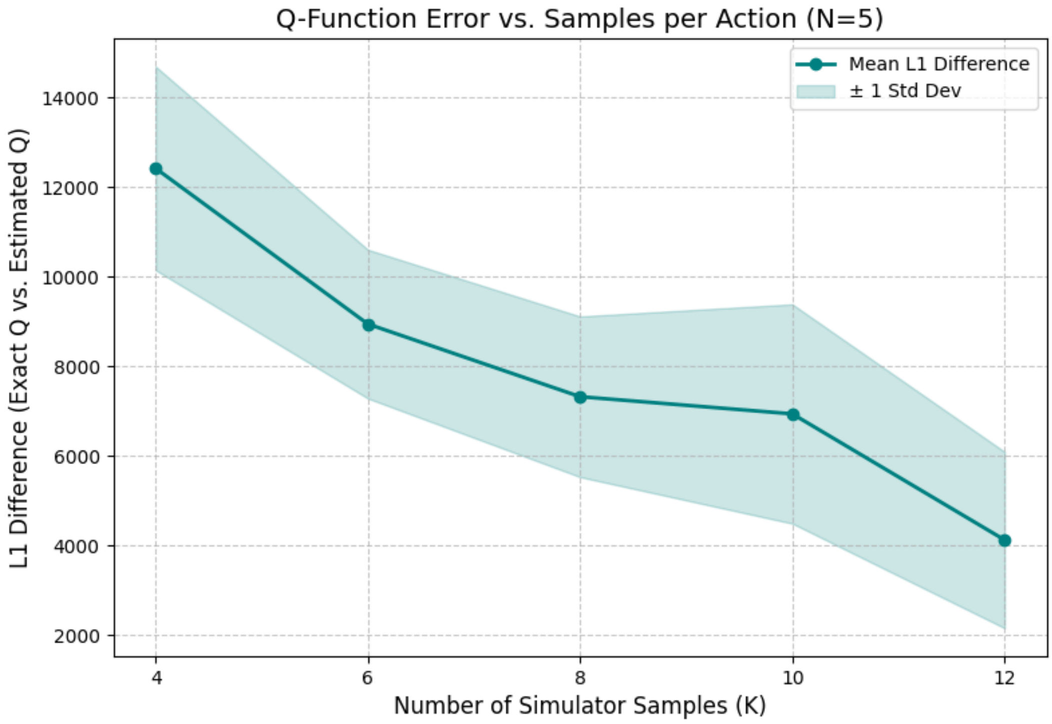
(c) Analysis using multiple seeds.- For each K:

1. Run experiment with multiple seeds
2. Compute L1 difference for each run
3. Calculate:
   - Mean error
   - Standard deviation

Mean Error $\text{Mean} = \frac{1}{M}\sum_{i=1}^{M} L\,1_i$

Standard Deviation- $\sigma = \sqrt{\frac{1}{M}\sum(L1_i - \text{Mean})^2}$

Plot of Mean L1 difference and Standard Deviation vs no of samples.



Observations-

1. P^→P as K→∞, ie estimated kernel becomes more accurate
2. Error decreases with K.
3. Variance reduces on increasing K.
4. Increasing K increases accuracy, but computation also increases.